

Eya Adouni

Research Master's in Information and Communication Sciences and Technologies

Vehicular networks (V2X), relay drones, machine learning · Full-stack developer

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Profile

Research Master's in ICT · Developer. Thesis carried out with LaBRI (Bordeaux) on relay-drone-assisted V2X communications: modelling, machine learning, decision platform. Seeking a PhD position in intelligent networks, applied AI, embedded systems, cybersecurity, or more broadly any innovative technology-driven field.

Research

Optimising V2X communications in coverage gaps through intelligent management of drone traffic

ENET'Com Sfax · LaBRI / ENSEIRB-MATMECA, Bordeaux · 2025 – 2026 · Supervisors: Ms Hend Marouane, Prof. Mohamed Mosbah, Mr Royston Fernandes · Scientific paper in preparation

- **Mathematical and physical modelling:** coverage geometry, optimal altitude under the regulatory ceiling, fleet sizing, Hungarian-algorithm assignment.
- **Operational Trust Index (OTI):** mission-dependent weights learned by random forest, non-compensatory safety factor, hysteresis, explainability.
- **Data and validation:** NASA Battery (SOH), DronePropA, RflyMAD · 8,000 profiles · Random Forest / XGBoost / LightGBM · Monte-Carlo 100% · SUMO simulation: 30% → 80% → 100% of vehicles connected.
- **Decision platform:** React / TypeScript, real map of Bordeaux, explained selection, digital twin, real-time log.

Experience

1 Jun – 24 Jul 2026	Research intern — LaBRI / ENSEIRB-MATMECA — Bordeaux, France Research internship funded by LaBRI: design, implementation and validation of the OTI and of the decision platform.
Jan – Jun 2024	Full-stack developer (final-year internship) — Amitech Smart Solutions — Remote, France AlloProf, educational social network with parental control (React, Node.js, MongoDB). Best final-year project 2024, distinguished by the Ministry.
since Jul 2022	IT officer — GAT Assurances — El Hamma, Gabès Customer records, IT incidents, cloud backups, remote support; internal customer-management platform.
Jan – Feb 2023	Back-end developer (internship) — Cyber Parc Djerba — Djerba, Tunisia Jobini, recruitment platform (PHP, Laravel, MySQL).
Jan – Feb 2022	Web developer (internship) — T&M Consulting — Tunisia Franchise Restaurant showcase website built with WordPress.

Education

2024 – 2026	Research Master's in Information and Communication Technologies — ENET'Com Sfax Next-generation networks, signal processing, embedded systems, cybersecurity. Ranked among the top of the cohort.
2021 – 2024	Bachelor's — Information Systems Development — ISET Djerba Graduated with honours. Best final-year project 2024.
2021	Baccalaureate in Experimental Sciences — Tahar Haddad High School, El Hamma

Languages

Arabic	Native
French	Fluent
English	Professional

Beyond work

Astronomy and space science, self-taught since childhood; small technical projects around astronomy (Code & Cosmos).

Skills

Machine learning
Python, scikit-learn, Random Forest, XGBoost, LightGBM, YOLOv8, reinforcement learning (RL), Monte-Carlo validation, data analysis
Networks and simulation
V2X, relay drones, coverage modelling, SUMO (TraCI, netconvert), signal processing
Development
React / TypeScript, Node.js, MERN, PHP / Laravel, Flutter, Leaflet, MySQL, MongoDB, Git
Security and tools
Cybersecurity fundamentals (Cisco, Fortinet), LaTeX, Office

Involvement

IEEE Student Branch, ISET Djerba (2022–2024)
General Secretary and trainer: JavaScript, cybersecurity and AI-basics workshops; organisation of events and hackathons, partnerships and sponsors. IEEE Young Professionals member.
4C Centre ambassador
Career and skills-certification centre, 2022.

Awards

Winners of TSYP 2024 (IEEE), Computer Science Department
Best final-year project 2024, distinguished by the Ministry
Entrepreneurial Hackathon SANAD CEED / HILFSWERK: 2nd place (2022)
SDG Hackathon, Polytech INTL: 3rd place
IoT Weekend Hackathon, ISET Djerba (2023) · Innovation Camp InJaz Tunisia (2021, 2022, 2023)

Certifications

- NASA Open Science 101 · 13 Aug 2026
- NSE 1 — Network Security Associate · 12 Apr 2026
- Introduction to Cybersecurity · 13 May 2026
- Intro to Machine Learning · 12 Apr 2026